



सत्यमेव जयते



पेटेंट कार्यालय, भारत सरकार

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पेटेंट प्रमाण पत्र

Patent Certificate

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पेटेंटी / Patentee

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प्रमाणित किया जाता है कि पेटेंटी को, उपरोक्त आवेदन में यथाप्रकरित **MULTI MATERIAL STRUCTURE WITH CONTROLLABLE MULTI DIRECTIONAL PROPERTY** नामक आविष्कार के लिए, पेटेंट अधिनियम, 1970 के उपबंधों के अनुसार आज तारीख अगस्त 2017 के पहले दिन से बीस वर्ष की अवधि के लिए पेटेंट अनुदत्त किया गया है।

It is hereby certified that a patent has been granted to the patentee for an invention entitled **MULTI MATERIAL STRUCTURE WITH CONTROLLABLE MULTI DIRECTIONAL PROPERTY** as disclosed in the above mentioned application for the term of 20 years from the 1st day of August 2017 in accordance with the provisions of the Patents Act, 1970.



[Signature]

Controller of Patents

अनुदान की तारीख : 31/01/2024
Date of Grant :

टिप्पणी - इस पेटेंट के नवीकरण के लिए फीस, यदि इसे बनाए रखा जाना है, अगस्त 2019 के पहले दिन को और उसके पश्चात प्रत्येक वर्ष में उसी दिन देय होगी।

Note. - The fees for renewal of this patent, if it is to be maintained, will fall / has fallen due on 1st day of August 2019 and on the same day in every year thereafter.

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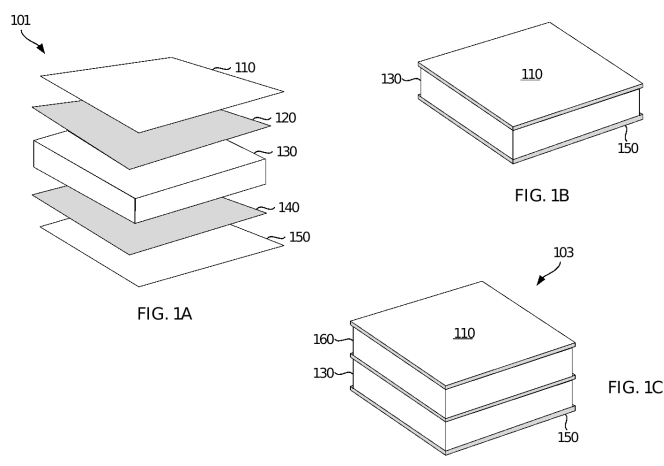
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(54) Title: MULTI MATERIAL STRUCTURE WITH CONTROLLABLE MULTI DIRECTIONAL PROPERTY

(57) Abstract: A multi-material structure comprising a first material with plurality of voids providing a first property in a first direction; and a second material possessing a second property disposed within the voids such that the structure exhibit the second property in a second direction while the first property is retained in the first direction. The voids of the first material are organized in the second direction and comprise at least one of circular, triangular, square and regular hexagonal (honeycomb) shape with at least 1mm dimension. The first material comprises at least one of plastic, aluminum, copper, cobalt, bismuth, bronze, chromium, gold, silver, iron, magnesium and titanium. The first property of the first material comprises at least one static property that does not alter with an application of external stimuli. The second property of the second material comprises at least one inherent property that alters dynamically with an application of external stimuli.



Form 2

The Patent Act 1970

(39 of 1970)

AND

Patent Rules 2003

Complete Specification

(Sec 10 and Rule 13)

Title	Multi Material Structure with Controllable Multi Directional Property
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The following specification particularly describes the invention and the manner in which it is to be performed.

DESCRIPTION

FIELD OF INVENTION

[0001] Embodiments of the present disclosure relate generally to smart structures comprising multiple materials and more specifically to a multi material structure with controllable multidirectional mechanical properties.

RELATED ART

[0002] Smart materials are the materials capable of changing their inherent properties such as shapes, dynamic properties (for example, stiffness, damping, thermal conductivity, magnetic permeability, etc.), or internal electrical properties including dielectric constant and resistivity depending on the changes in their environment or by supplying external stimuli.

[0003] Various types of smart materials for example, piezoelectric materials, magnetostrictive materials, rheological fluids, shape memory alloys, thermoelectric materials and smart gels are being used for various applications by military, aerospace, construction industries, smart packaging services and others.

[0004] A structure developed by a homogenous material is not able to fulfill entire required properties for desired applications. For example, a material having higher stiffness may not possess required damping and vice versa to achieve desired vibration in a system. To overcome this deficiency, multi-material structures are developed in the form of layered structures using the smart materials. For example, the smart materials with damping properties are sandwiched between thin layers of identical stiff materials of the structure by depositing layer by layer to isolate vibrations/noise from a system.

[0005] Conventionally, the smart materials in the sandwich structure acts as an intermediate between the two identical stiff structures and are stacked together using an adhesive to absorb vibrations/noise administered onto it. For example, the damping structures formed by using the smart materials are highly compatible with substantially all apparatus and yields superior energy absorption compared to conventional damping configurations without any smart material.

[0006] However, conventional structures developed by using the smart materials are inefficient in achieving desired multiple properties. For example, the damping structures with a smart material are not effective in isolating vibrations in a system because of poor bonding between the layer by layer deposited stacks of smart materials and stiff materials. Application of high shear force onto the structure results in disruption or unwrapping of the smart material from the stacks of layers

which in turn result in failure of entire damping system. Hence, a multi-material structure with controllable multidirectional properties that overcomes the aforesaid limitations is required.

SUMMARY

[0007] According to an aspect of the present disclosure, a multi-material structure comprising a first material with plurality of voids providing a first property in a first direction; and a second material possessing a second property disposed within the voids such that the structure exhibit the second property in a second direction while the first property is retained in the first direction. The voids of the first material are organized in the second direction (i.e., along a direction of load) and comprise at least one of circular, triangular, square and regular hexagonal (honeycomb) shape with at least 1mm dimension. The first material comprises at least one of plastic, aluminum, copper, cobalt, bismuth, bronze, chromium, gold, silver, iron, magnesium and titanium.

[0008] The first property of the first material comprises at least one static property that does not alter with an application of external stimuli. The second property of the second material comprises at least one inherent property that alters dynamically with an application of external stimuli. The second material comprises at least one of magneto rheological fluid (MRF), magneto rheological elastomer (MRE), magneto rheological grease (MRG), piezo-material, electro rheological fluid (ERF), magnetostrictive materials and shape memory alloys (SMA).

[0009] Several aspects are described below, with reference to diagrams. It should be understood that numerous specific details, relationships, and methods are set forth to provide a full understanding of the present disclosure. One who skilled in the relevant art, however, will readily recognize that the present disclosure can be practiced without one or more of the specific details, or with other methods, etc. In other instances, well-known structures or operations are not shown in detail to avoid obscuring the features of the present disclosure.

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG.1A through FIG. 1C are the diagrams illustrating conventional sandwich structure of an isolation system using smart materials.

[0011] FIG. 2A and FIG. 2B are the diagrams illustrating a multi-material structure for achieving multidirectional properties in an embodiment of the present disclosure.

[0012] FIG. 3A through FIG. 3D are the diagrams illustrating various geometrical shapes and sizes of the voids in a primary material of the present disclosure.

[0013] FIG. 4 is a flowchart illustrating the steps involved in achieving multidirectional properties of a multi-material using smart materials in an embodiment of the present disclosure.

[0014] FIG. 5 is a diagram illustrating a multi-material composite structure using at least two different smart materials in an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EXAMPLES

[0015] FIG.1A through FIG. 1C are the diagrams illustrating conventional sandwich structure of an isolation system using smart materials. In recent days, using smart materials in various systems and/or apparatus has significantly increased due to achievement of controllable properties by external stimuli. Using smart material structures, the desired properties like thermal, damping, magnetic properties etc, are able to modify as per the requirement by using an external stimuli. As shown in the FIG. 1A, 101 represents a smart sandwich structure comprising a top sheet 110, an adhesive layer (120 & 140), a smart material 130 and a bottom sheet 150.

[0016] Application of unbalanced external force (for example, shear force) onto this mechanical structure results in vibrations and/or noise which have a direct impact on a system. In order to avoid the vibrations/noise in a system, smart materials having the inherent properties of isolating vibrations/noise are introduced between the top sheet 110 and the bottom sheet 150. As shown in the FIG. 1B, the smart material 130 is kept in contact with the top sheet 110 and the bottom sheet 150 by using a couple of adhesive layers of shape and size identical to that of the structure 101. As the smart material 130 is sandwiched between the top sheet 110 and the bottom sheet 150, the vibrations or noise directed onto the structure 101 gets reduced or cancelled by the inherent property of the smart material 130 used in the structure by using external stimuli.

[0017] FIG. 1C is a diagram illustrating a conventional composite structure comprising plurality of smart materials. As shown there, the structure 103 comprises two smart materials 130 and 160 having different inherent properties that are sandwiched between the face sheet 110 and the supporting structure 150 using adhesive layers 120 and 140. These composite structures are useful in achieving multidirectional properties with multi-materials as desired.

[0018] Also, the smart material 130 and/160 gets slipped away/misaligned from the structure 101 and/or 103 on application of a continuous shear force onto it. This misalignment of the smart material from its structure has a direct effect on control efficiency and costs of the entire system. Hence, a multi-material structure with multidirectional properties is provided in the present disclosure and is further discussed in the following figures.

[0019] FIG. 2A and FIG. 2B are the diagrams illustrating a multi-material structure for achieving multidirectional properties in an embodiment of the present disclosure. As shown there, the multi-material structure of the present disclosure comprises a primary material and a secondary material wherein the secondary material is filled into plurality of voids that are created within the primary material. In an embodiment, the primary material comprises any structure of a system having a mechanical property to perform a specific action in a desired way whereas the secondary material comprises a smart material having inherent properties that alter by an external stimulus.

[0020] The smart materials comprise at least one of magneto rheological fluid (MRF), magneto rheological elastomer (MRE), magneto rheological grease (MRG), piezo-material, electro rheological fluid (ERF), magnetostrictive materials, shape memory alloys (SMA) etc. By filling the voids in the primary material with the secondary material, static and dynamic behavior/properties of the resulting structure are controlled by using external stimuli such as electrical voltage/current, magnetic field, temperature, humidity, light, sound and the like.

[0021] In another embodiment, the voids created in the primary material are of different shape, size and in any combination thereof. For example, the voids created in the primary structure are having any of a geometrical shape and size such as circle, triangle, square, hexagon (honeycomb), etc., and any combination of these shapes. The voids in the primary material are achieved by using conventional mechanical means such as piercing, casting, layer by layer manufacturing or any other manufacturing process.

[0022] The voids in the primary material are provided in such a way that structural properties of the primary material remain same in a desired direction i.e., a primary material with structural property such as stiffness is retained in a desired direction by creating honeycomb (hexagon) voids in it. In these voids, a secondary material comprising a smart material such as MRE having controlled damping effect is filled uniformly to achieve both damping and stiffness for the multi-material structure.

[0023] FIG. 3A through 3D are the diagrams illustrating various geometrical shapes and sizes of the voids in a primary material of the present disclosure. As discussed in the previous embodiments, the voids in the primary material of the multi-material structure comprises of various geometrical shapes and sizes. As shown in the FIG. 3A, the multi-material structure 301 comprises a primary material 310 and a secondary material 320 filled in the circular voids 330 of the primary material 310.

[0024] In an example, the primary material 310 comprises a mechanical structure with a substantial rigidity in a first direction as its physical property whereas the secondary material 320 comprises a smart material with inherent damping, noise, thermal, magnetic properties etc. filled in the circular voids 330 of the primary material 310. The voids 330 are created in such a manner that physical properties of the primary material 310 are retained within the structure 301 along with the secondary material 320. This results in the multi-material structure 301 comprising both physical property of the primary material 310 as well as the inherent property drawn from the secondary material 320.

[0025] FIG. 3B is a diagram illustrating the multi-material structure with honeycomb shaped (hexagonal) voids in the primary material 310 with a principle property and a secondary material comprising a desired smart material of required property. The honeycomb shaped voids helps in retaining the principle property of the primary material in a desired direction and further these honeycomb shaped voids are compatible with majority of the applications. Further, FIG. 3C and FIG. 3D are the diagrams illustrating the multi-material structures with rectilinear shaped voids and triangular truss shaped voids in a primary material 310 respectively wherein the secondary material comprising a desired smart material is filled in the voids to achieve multidirectional properties for the multi-material structure.

[0026] FIG. 4 is a flowchart illustrating the steps involved in achieving multidirectional properties of a multi-material using smart materials in an embodiment of the present disclosure. In step 410, a primary structure of desired physical property is selected from conventional structures. The primary structure may comprise at least one of plastic (ABS, PP, etc.), aluminum, copper, cobalt, bismuth, bronze, chromium, gold, silver, iron, magnesium, titanium and the like. These materials are being used for various applications in various manufacturing and construction industries but not limited to aerospace machinery/equipment.

[0027] In step 420, a desired geometrical shape and size of voids are created in the primary material by conventional mechanical means of any manufacturing process. In an embodiment, the primary material is developed by layer by layer deposition of the primary material along with voids of desired shape and size which avoids the step 420. In an example, honeycomb (regular hexagonal) shaped voids of size 1mm dimension is provided in the primary structure in such a way that physical properties or strength of the primary material is not compromised. In an

embodiment, the voids in the primary material may vary from shape and size as per the requirement.

[0028] In step 430, the voids in the primary material are filled with a smart material of desired property to induce the desired property of the smart material into the primary material. In an example, the smart materials comprises any material capable of changing its inherent properties responding to an external stimuli such as electrical voltage/current, magnetic field, temperature, humidity, light and the like.

[0029] In step 440, the primary material inscribed with the smart material in the voids possesses structural strength or physical properties of the primary material as well as the inherent property of the smart material. Thus, a multi-material structure is developed using different materials and multidirectional properties (in any direction as desired) are achieved by creating and filling the voids in the primary structure with the smart material. Further, number of voids in the primary structure filled with the smart material controls the multidirectional properties of the multi-material structure using the external stimuli.

[0030] FIG. 5 is a diagram illustrating a multi-material composite structure using at least two different smart materials in an embodiment of the present disclosure. As discussed in the previous embodiment, the multi-material structure of the present disclosure comprises a smart material filled in the voids of the primary material achieving multidirectional properties using external stimuli.

[0031] However, the smart material properties are not required throughout the multi-material structure and further it may require different properties (other than the property of the primary material and the smart material used) for different purposes. In such cases, a composite multi-material structure having various smart materials is disclosed in the present embodiment. As shown in the FIG. 5, the multi-material structure comprises a primary material with voids of varying geometrical shapes and sizes.

[0032] Further, different smart materials are incorporated into the voids of the primary material so that a desired property of the smart material is achieved in a desired direction without disturbing the principle properties of the primary material using the external stimuli. Thus, the multi-material structure of the present disclosure helps in achieving controlled multi directional property by embedding secondary material into the voids of primary material.

[0033] While various embodiments of the present disclosure have been described above, it should be understood that they have been presented by way of example only, and not limitation. Thus, the breadth and scope of the present disclosure should not be limited by any of the above-discussed embodiments, but should be defined only in accordance with the following claims and their equivalents.

CLAIMS

I/We Claim,

1. A multi-material structure comprising:
a first material with plurality of voids providing a first property in a first direction; and
a second material possessing a second property disposed within the voids such that the structure exhibit the second property in a second direction while the first property is retained in the first direction.
2. The multi-material structure of claim 1, wherein the voids of the first material are organized in the second direction and comprise at least one of circular, triangular, square and regular hexagonal (honeycomb) shape with at least 1mm dimension.
3. The multi-material structure of claim 2, wherein the first material comprises at least one of plastic, aluminum, copper, cobalt, bismuth, bronze, chromium, gold, silver, iron, magnesium and titanium.
4. The multi-material structure of claim 3, wherein the first property of the first material comprises at least one static property that does not alter with an application of external stimuli.
5. The multi-material structure of claim 1, wherein the second property of the second material comprises at least one inherent property that alters dynamically with an application of external stimuli.
6. The multi-material structure of claim 5, wherein the external stimuli comprises at least one of electrical voltage/current, magnetic field, temperature, humidity, light and sound.
7. The multi-material structure of claim 5, wherein the second material comprises at least one of magneto rheological fluid (MRF), magneto rheological elastomer (MRE), magneto rheological grease (MRG), peizo-material, electro rheological fluid (ERF), magnetostrictive materials and shape memory alloys (SMA).
8. A method, system and apparatus providing one or more features as described in the paragraphs of this specification.

Date: 01-08-2017

Signature.....

ABSTRACT

A multi-material structure comprising a first material with plurality of voids providing a first property in a first direction; and a second material possessing a second property disposed within the voids such that the structure exhibit the second property in a second direction while the first property is retained in the first direction. The voids of the first material are organized in the second direction and comprise at least one of circular, triangular, square and regular hexagonal (honeycomb) shape with at least 1mm dimension. The first material comprises at least one of plastic, aluminum, copper, cobalt, bismuth, bronze, chromium, gold, silver, iron, magnesium and titanium. The first property of the first material comprises at least one static property that does not alter with an application of external stimuli. The second property of the second material comprises at least one inherent property that alters dynamically with an application of external stimuli.

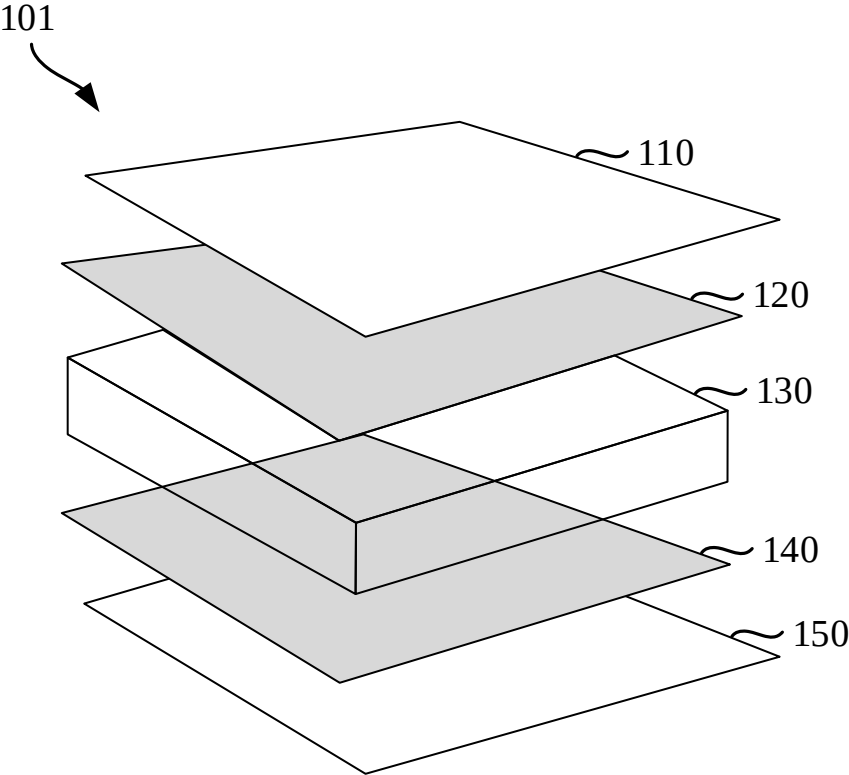


FIG. 1A

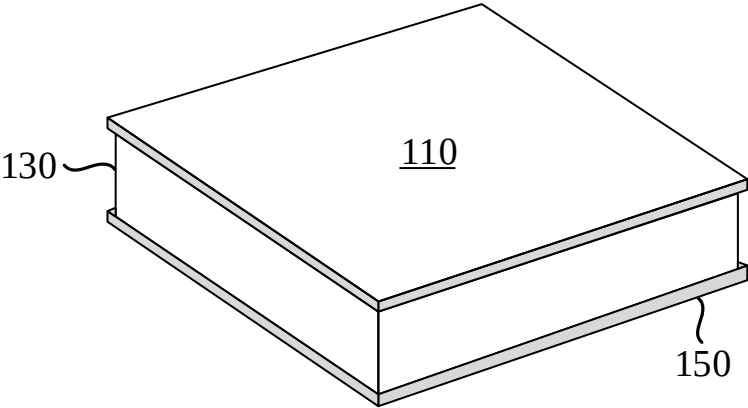


FIG. 1B

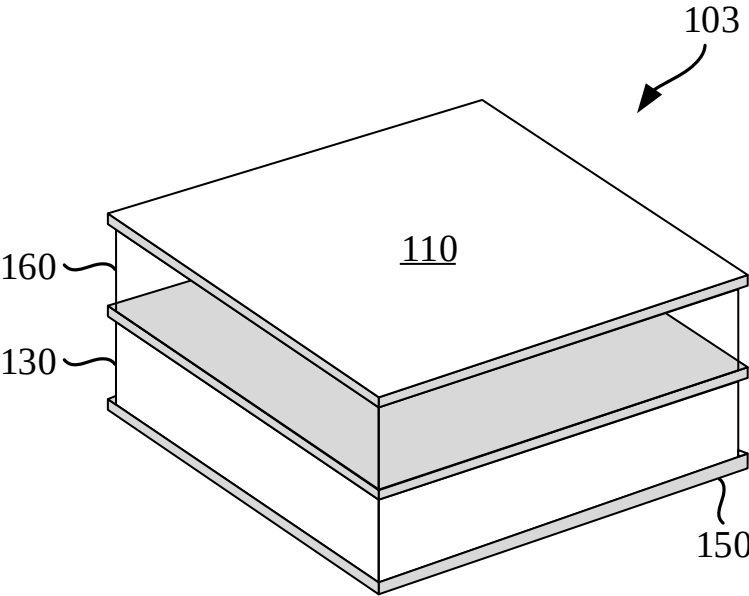


FIG. 1C

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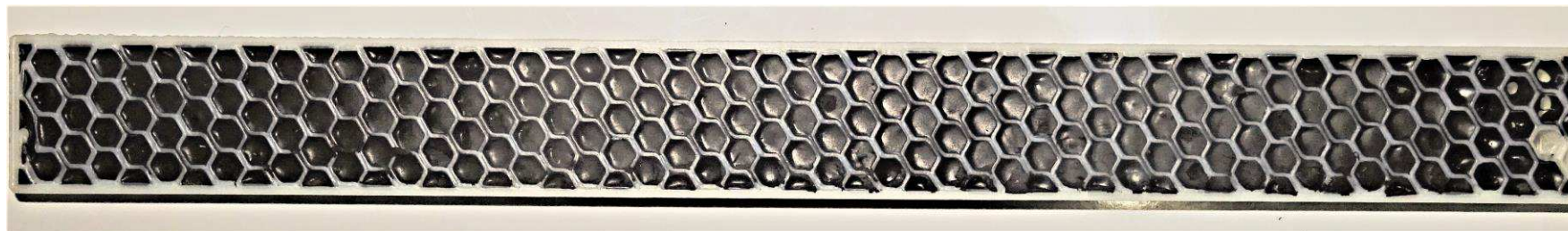


FIG. 2A

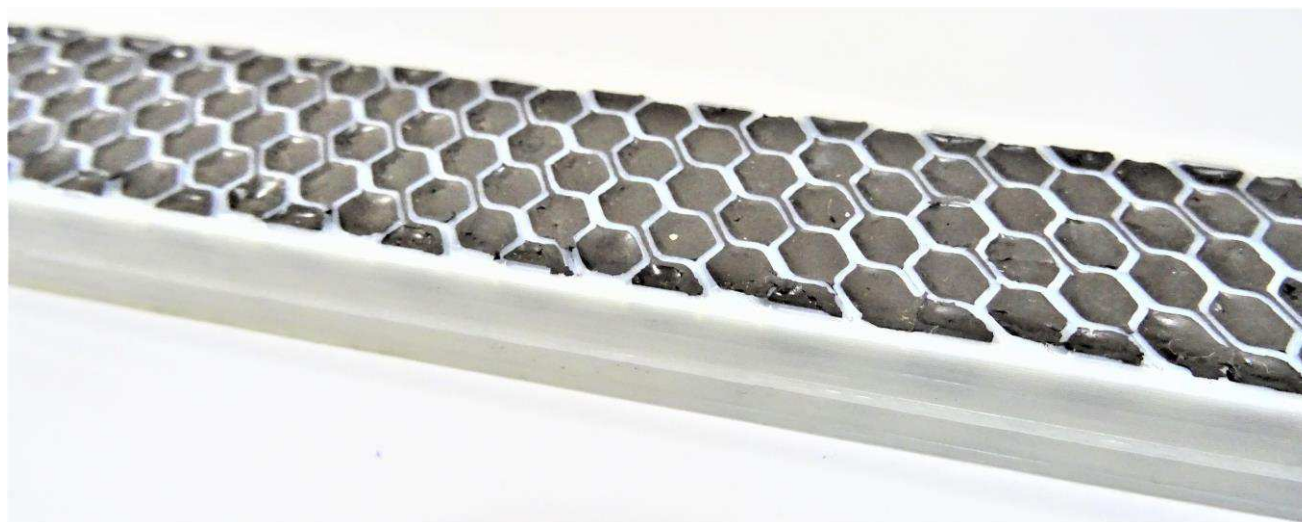


FIG. 2B

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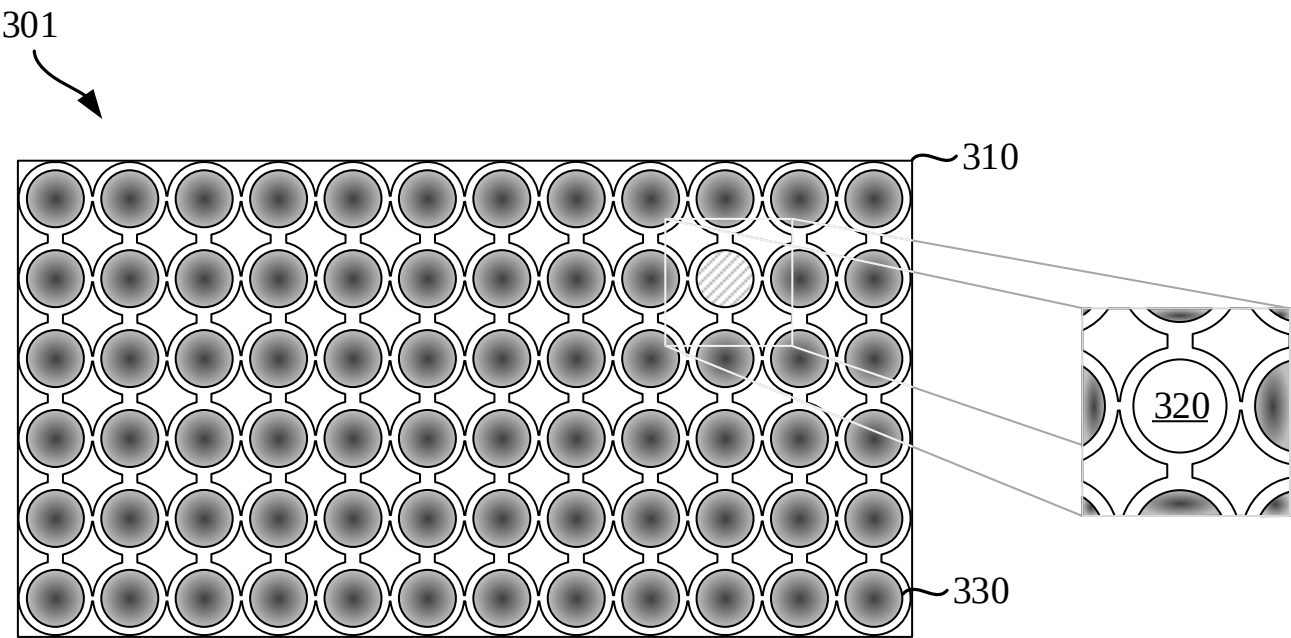


FIG. 3A

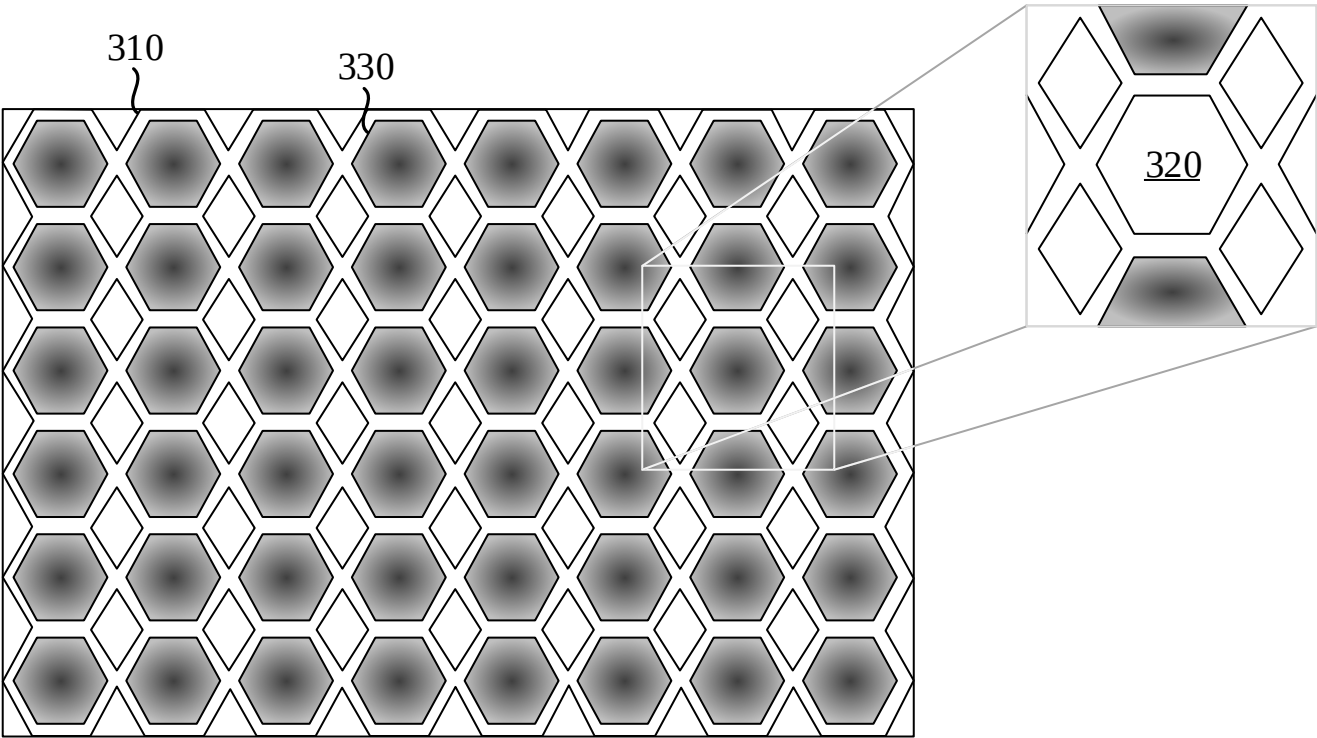


FIG. 3B

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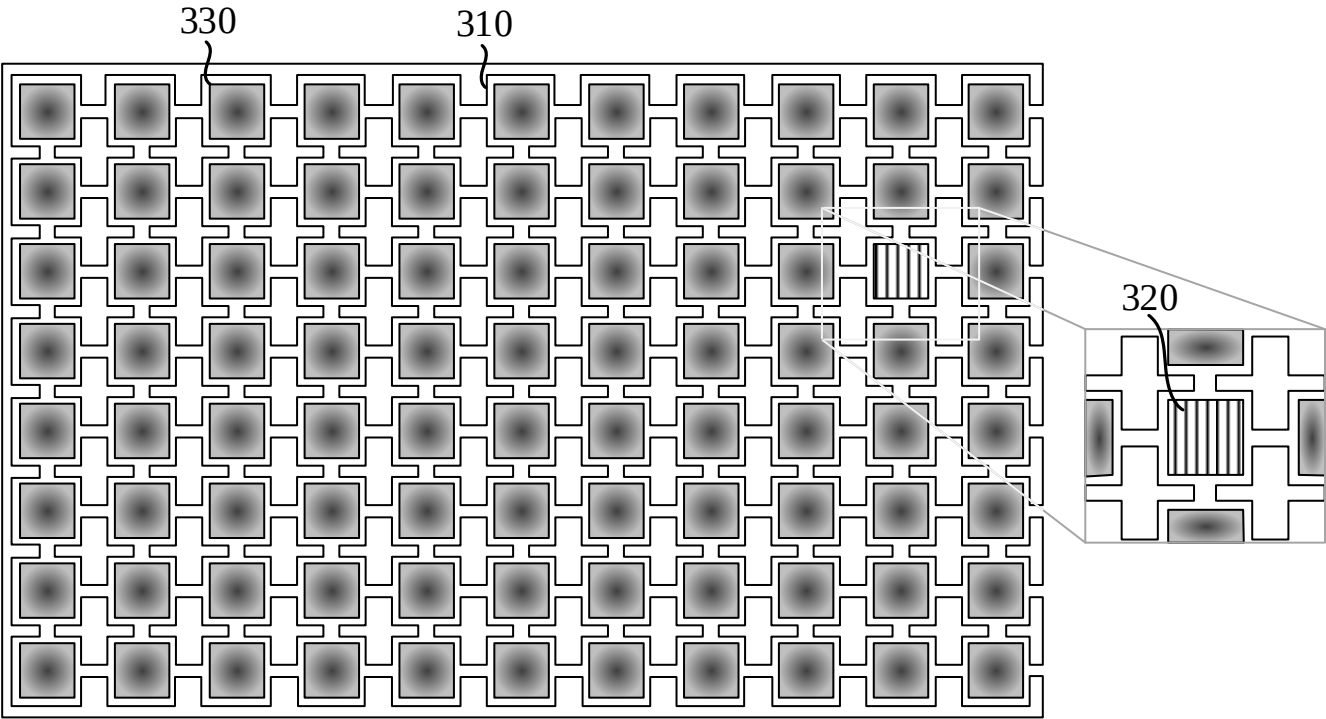


FIG. 3C

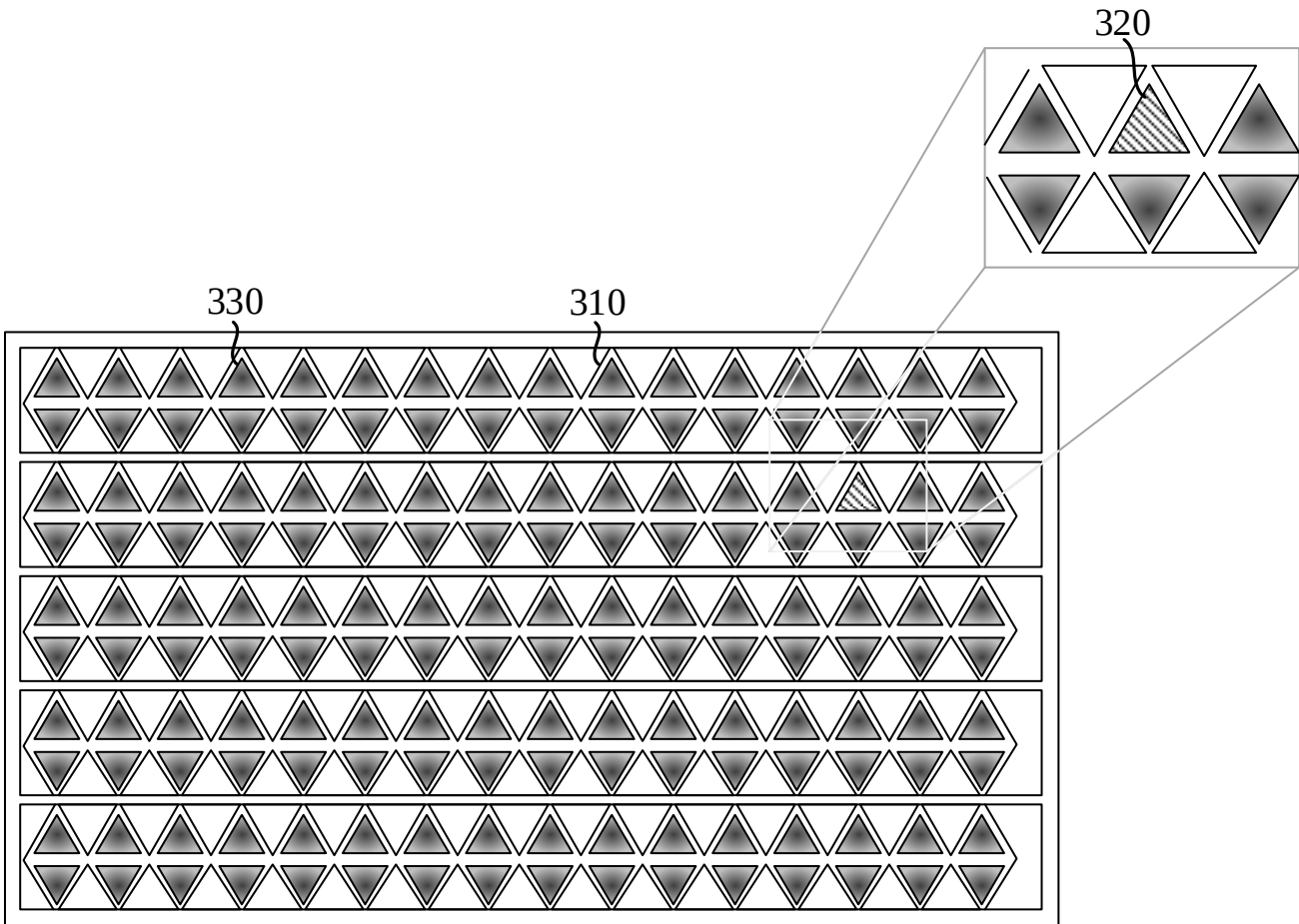


FIG. 3D

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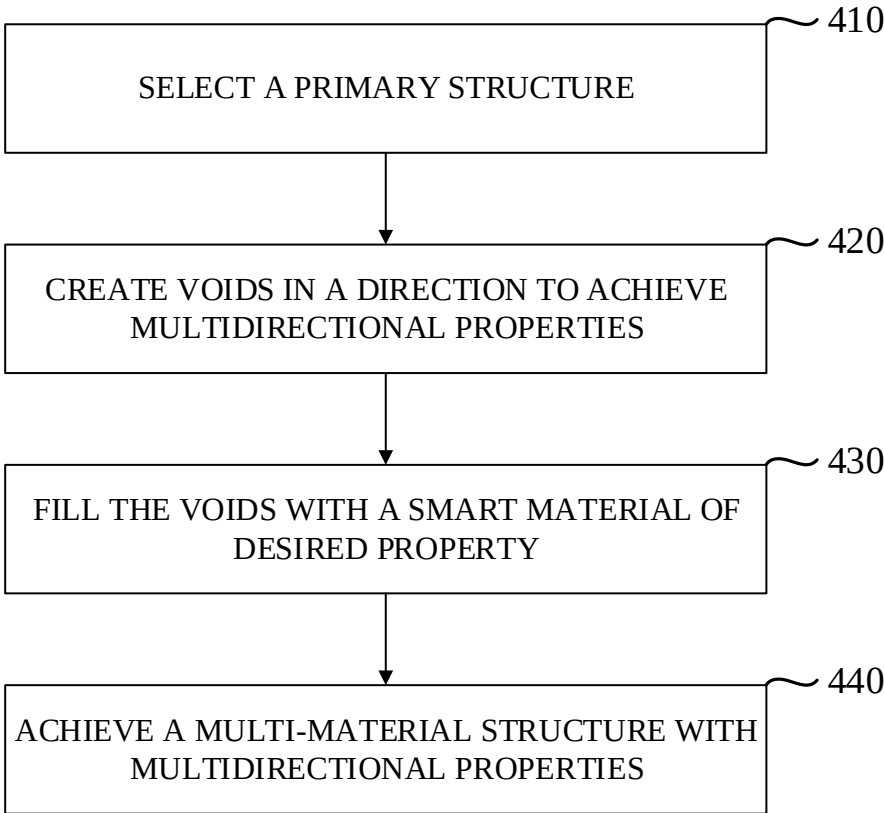


FIG. 4

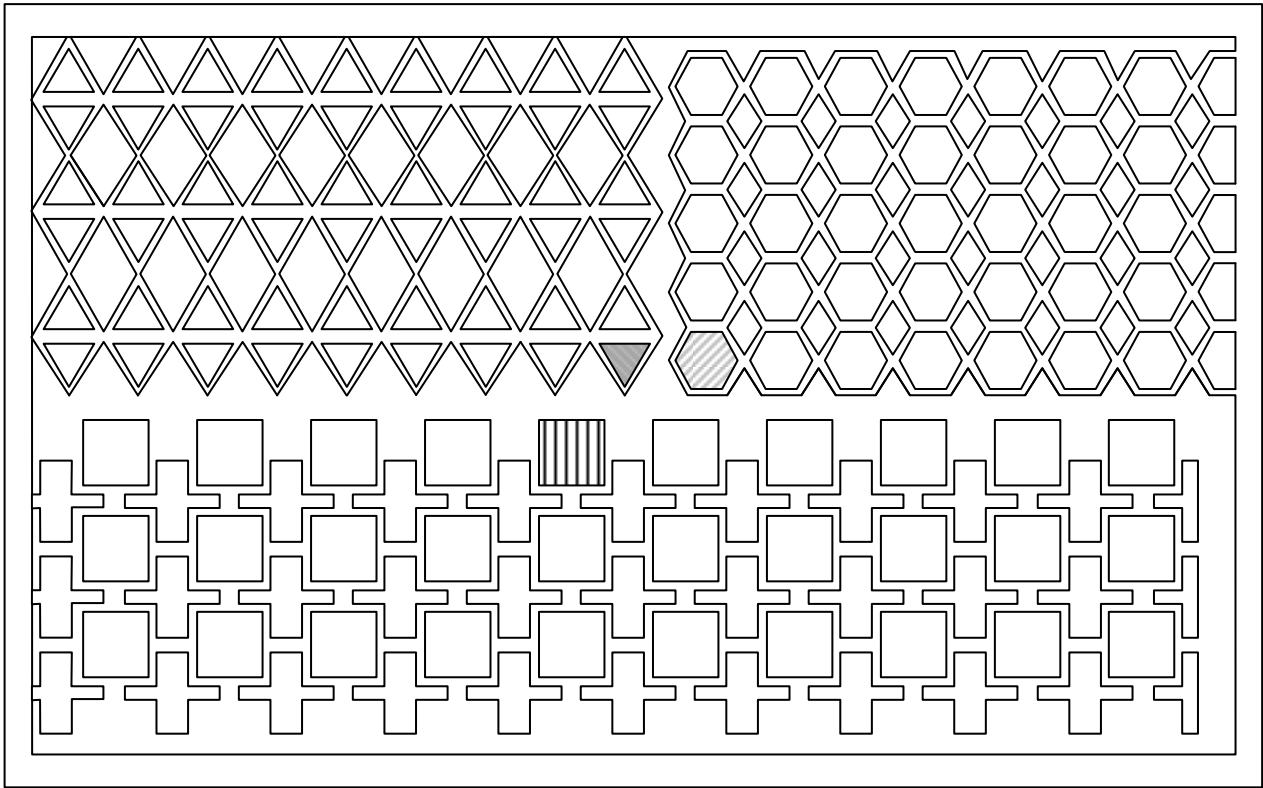


FIG. 5

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